

EMSID recap

The next slide repeats the EMSID recipe from page 29 of the handout `2c1_linear_equations.pdf` for convenient reference.

Solving linear 1st order ODEs

Nonhomogeneous first order linear equation in standard form:

$$\frac{dy}{dx} + P(x)y = f(x) \quad (*)$$

EMSID *method to find the general solution:*

- 1 **E**valuate the integrating factor $e^{\int P(x) dx}$
- 2 **M**ultiply both sides of (*) by the integrating factor
- 3 **S**implify using product rule:

$$\frac{d}{dx} \left(e^{\int P(x) dx} y \right) = e^{\int P(x) dx} f(x) \quad (**)$$

- 4 **I**ntegrate both sides of (**) (but add only one constant).
- 5 **D**ivide both sides by the integrating factor (to isolate y).

Note: "EMSID" isn't standard terminology; it's just a suggestion to help you remember the method.

1 Other First Order Methods

- Announcements
- Bernoulli equations
- Homogeneous nonlinear equations
- Exact equations
- Integrating factors for differential forms
- Practice problems



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

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Announcements

- The TA is Illya Kierkosz (kierkosi@mcmaster.ca)
 - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: <https://davidearn.github.io/math2c/>
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on [childsmath](#).
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
 - Subject: Math 2C03: confusion about assignment 1
 - Subject: Math 2C03: conflict with test window
- Office hour: Thursdays, 9:30–10:20 am, after class.

Test 1: action required

TEST 1

ACTION REQUIRED

[Course tests page](#)

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Bernoulli equations

Bernoulli equations (♡)

A *Bernoulli equation* has the form

$$\frac{dy}{dx} + p(x)y = f(x)y^r, \quad r \neq 0, 1. \quad (\heartsuit)$$

The excluded cases $r = 0$ and $r = 1$ reduce to *first-order linear ODEs*, which we have already solved using EMSID. If $f \neq 0$, the term $f(x)y^r$ makes the equation nonlinear.

Replacing the RHS by zero gives the *complementary linear equation*:

$$\frac{dy}{dx} + p(x)y = 0,$$

As before, find a nonzero solution y_1 of this equation, then try

$$y = u(x)y_1(x),$$

i.e., let's try *the same variation of parameters technique*.

Bernoulli equations (♡)

Substitute $y = uy_1$ into $\frac{dy}{dx} + p(x)y = f(x)y^r$:

$$y_1 \frac{du}{dx} + u \frac{dy_1}{dx} + p(x)uy_1 = f(x)(uy_1)^r$$

$$y_1 \frac{du}{dx} = f(x)y_1^r u^r \quad (*)$$

since $\frac{dy_1}{dx} + p(x)y_1 = 0$.

For $u \neq 0$, this is *separable*:

$$u^{-r} \frac{du}{dx} = f(x)y_1^{r-1}.$$

Before dividing by u^r : check whether $u \equiv 0$, hence $y \equiv 0$, is also a solution.

Bernoulli equations (♡)

Example (Solve $\frac{dy}{dx} - y = xy^2$)

The **complementary linear equation** is $\frac{dy}{dx} - y = 0$. By **EMSID**, $P(x) = -1$, so the **integrating factor** is e^{-x} . Thus $\frac{d}{dx}(e^{-x}y) = 0$ gives $y = ce^x$. Take $y_1 = e^x$.

Set $y = ue^x$. The **ODE (*)** for u on the preceding slide becomes

$$\begin{aligned}e^x \frac{du}{dx} &= xu^2 e^{2x} \\ \frac{1}{u^2} \frac{du}{dx} &= xe^x, \quad (u \neq 0). \\ -\frac{1}{u} &= \int xe^x dx = (x-1)e^x + C.\end{aligned}$$

Bernoulli equations (♡)

Example (Solve $\frac{dy}{dx} - y = xy^2$)

Rearranging the result from the preceding slide,

$$\frac{1}{u} = (1 - x)e^x - C, \quad y = ue^x = \frac{1}{1 - x - Ce^{-x}}.$$

This formula applies on intervals where its denominator is nonzero.

We divided by u^2 . The solution $u \equiv 0$ was therefore excluded from the calculation. It gives the additional solution

$$y(x) \equiv 0.$$

Bernoulli equations (♡)

Alternative: transform (♡) into a *linear* ODE, then use EMSID. First note that if $r > 0$ then $y \equiv 0$ solves (♡). Then assume $y \neq 0$. (*Also note:* if $r \notin \mathbb{Z}$ then y^r may not be real for all $y \in \mathbb{R}$, in which case assume $y > 0$.) Then set:

$$v = y^{1-r} \quad (*)$$

Since

$$\frac{dv}{dx} = (1-r)y^{-r} \frac{dy}{dx},$$

multiplying the Bernoulli equation (♡) by $(1-r)y^{-r}$ gives

$$\frac{dv}{dx} + (1-r)p(x)v = (1-r)f(x).$$

Solve this *linear ODE for v* by EMSID, then recover y from (*), i.e., $y = v^{1/(1-r)}$.

Bernoulli equations (♡)

Example (Solve IVP $\frac{dy}{dx} - y = xy^2$, $y(0) = 2$)

This is a Bernoulli equation (♡) with $r = 2$, $p(x) = -1$, and $f(x) = x$.

- $y \equiv 0$ solves $\frac{dy}{dx} - y = xy^2$, but not $y(0) = 2$, so assume $y \neq 0$.
- For $y \neq 0$, set $v = y^{-1} = \frac{1}{y}$. The linear ODE for v is

$$\frac{dv}{dx} + v = -x.$$

By EMSID, $P(x) = 1$ and the integrating factor is e^x , so:

$$\frac{d}{dx}(e^x v) = -xe^x,$$

$$e^x v = (1 - x)e^x - C,$$

$$y = \frac{1}{v} = \frac{1}{1 - x - Ce^{-x}}, \quad 1 - x - Ce^{-x} \neq 0.$$

$$\text{Finally, } y(0) = 2 \implies 2 = \frac{1}{1 - C} \implies C = \frac{1}{2} \implies y = \frac{1}{1 - x - \frac{1}{2}e^{-x}}.$$

Homogeneous nonlinear equations

Homogeneous nonlinear equations

Trench (§2.4) uses another (common) definition of *homogeneous*: an ODE is homogeneous if scaling all variables by a non-zero constant leaves the ODE unchanged.

Typical first order homogeneous ODEs can be written

$$\frac{dy}{dx} = q\left(\frac{y}{x}\right) \quad \text{where } q \text{ is a function of one variable} \\ \text{(the quotient of } y \text{ and } x\text{).}$$

Multiplying x and y by the same nonzero constant leaves both sides of the equation unchanged.

This does *not* mean “homogeneous linear”, as in $\frac{dy}{dx} + p(x)y = 0$. A nonlinear homogeneous equation *need not be linear*.

Recognizing a homogeneous nonlinear equation

For a quotient of polynomials, divide numerator and denominator by the same power of x . For example, if $x \neq 0$,

$$\frac{dy}{dx} = \frac{x^2 + 3xy + y^2}{x^2 + 2xy} \iff \frac{dy}{dx} = \frac{1 + 3(y/x) + (y/x)^2}{1 + 2(y/x)}.$$

This has the form $\frac{dy}{dx} = q(y/x)$.

What if the numerator were $x^n + 3xy + y^2$? For which n would the same method work? *Only $n = 2$* , so every term has the same total degree before dividing by x^2 .

Solving homogeneous nonlinear equations

Substituting $y = xu$ transforms

$$\frac{dy}{dx} = q\left(\frac{y}{x}\right) \implies u + x \frac{du}{dx} = q(u).$$

Thus

$$x \frac{du}{dx} = q(u) - u.$$

This is a *separable equation*.

Before dividing by $q(u) - u$, check for constant solutions $u \equiv c$ with $q(c) = c$. Each gives a solution $y = cx$ of the original equation.

Solving homogeneous nonlinear equations

Example (Solve $\frac{dy}{dx} = \frac{y}{x} + \left(\frac{y}{x}\right)^2$, $x > 0$)

Here $q(u) = u + u^2$.

Set $y = xu \implies u + x \frac{du}{dx} = u + u^2 \implies x \frac{du}{dx} = u^2$

Note that $u \equiv 0$ is a solution. It gives $y \equiv 0$.

For $u \neq 0$, separate and integrate:

$$\begin{aligned} \frac{1}{u^2} \frac{du}{dx} &= \frac{1}{x} \implies -\frac{1}{u} = \ln x + C \\ \implies y &= xu = \frac{-x}{\ln x + C} \end{aligned}$$

on intervals where $\ln x + C \neq 0$,

i.e., where $x \neq e^{-C}$.

Exact equations

Exact equations

Write a first order equation in *differential form*:

$$M(x, y) dx + N(x, y) dy = 0, \quad (\spadesuit)$$

which does not commit us to which variable is dependent.

If $y = y(x)$, () means

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0.$$

Suppose we can find a function $\Phi(x, y)$ such that

$$\frac{\partial \Phi}{\partial x} = M, \quad \frac{\partial \Phi}{\partial y} = N.$$

Then we say the equation is *exact*.

Exact equations

If $\Phi_x = M$ and $\Phi_y = N$, the chain rule gives

$$\frac{d}{dx}\Phi(x, y(x)) = \Phi_x + \Phi_y \frac{dy}{dx} = M + N \frac{dy}{dx} = 0.$$

So the solutions are described *implicitly* by

$$\Phi(x, y) = C$$

The function Φ is called a *potential*.

If we are given an initial condition, then it determines C .

Note: (For those to whom elementary physics and vector calculus are familiar.) The vector field (M, N) is said to have *scalar potential* Φ if its gradient $\nabla\Phi = (\Phi_x, \Phi_y) = (M, N)$. If (M, N) represents a force, then $M dx + N dy$ gives the infinitesimal work done over a displacement (dx, dy) . Such a force is *conservative* (i.e., total mechanical energy is conserved if no other forces act): it can be written $-\nabla U$, where $U = -\Phi$ is potential energy.

Exactness test

How can we tell whether a potential Φ exists?

If $\Phi_x = M$ and $\Phi_y = N$, and Φ_{xy} and Φ_{yx} exist and are continuous, then

$$\frac{\partial M}{\partial y} = \Phi_{xy} = \Phi_{yx} = \frac{\partial N}{\partial x}.$$

Theorem (Exactness condition, [Trench §2.5](#))

If M , N , M_y , and N_x are continuous on an open rectangle R , then $M dx + N dy = 0$ is exact on R if and only if

$$M_y = N_x \quad \text{throughout } R.$$

Solving an exact equation

Example (Solve $(2xy + 1) dx + (x^2 + 2y) dy = 0$)

$M = 2xy + 1$ and $N = x^2 + 2y \implies M_y = 2x = N_x \implies$
the equation is exact.

Find Φ by integrating $\Phi_x = M$ with respect to x :

$$\Phi(x, y) = \int (2xy + 1) dx = x^2y + x + g(y).$$

The “constant” of integration may depend on y .

Now require $\Phi_y = N$:

$$x^2 + g'(y) = x^2 + 2y \implies g(y) = y^2.$$

Therefore

$$x^2y + x + y^2 = C.$$

If we specify the initial condition $y(0) = 1$, then $C = 1$.

Finding parameters that make an equation exact

Example ($(2xy + Ay^2 + By) dx + (x^2 + 6xy + 4x + 2y) dy = 0$)

For which A and B is the equation exact?

The condition $M_y = N_x$ becomes

$$2x + 2Ay + B = 2x + 6y + 4,$$

which must hold for *every* (x, y) in a rectangle R . So assume it holds for $(x, y) = (0, 0)$ ($\therefore B = 4$) and $(x, y) = (0, 1)$ ($\therefore A = 3$).

$$\therefore M = 2xy + 3y^2 + 4y, \quad N = x^2 + 6xy + 4x + 2y.$$

We can now solve the equation. Integrating $\Phi_x = M$ gives

$$\Phi = x^2y + 3xy^2 + 4xy + g(y).$$

Then $\Phi_y = N$ implies $g'(y) = 2y$. Thus an implicit solution is

$$x^2y + 3xy^2 + 4xy + y^2 = C.$$

If we specify $y(1) = 1$, then $C = 9$.

Integrating factors for differential forms

If $M dx + N dy = 0$ is *not exact*, sometimes multiplying by a function $\mu(x, y)$ makes it exact:

$$\mu M dx + \mu N dy = 0.$$

Such a function μ is an *integrating factor*.

We already used an integrating factor for a *linear* first order ODE. The present problem is more general: our starting equation may be nonlinear, and we want to make a differential form exact.

An integrating factor depending only on x

Suppose $\mu = \mu(x)$ and $N \neq 0$ on the region being considered. Exactness of the multiplied equation requires

$$(\mu M)_y = (\mu N)_x \quad \Longleftrightarrow \quad \mu M_y = \mu' N + \mu N_x.$$

Hence

$$\frac{\mu'}{\mu} = \frac{M_y - N_x}{N}.$$

If the right-hand side depends *only on x* , call it $p(x)$ and take

$$\mu(x) = e^{\int p(x) dx}.$$

An integrating factor depending only on y

Similarly, if $\mu = \mu(y)$ and $M \neq 0$, exactness requires

$$\mu' M + \mu M_y = \mu N_x.$$

Therefore

$$\frac{\mu'}{\mu} = \frac{N_x - M_y}{M}.$$

If this quotient depends *only on y* , call it $q(y)$ and take

$$\mu(y) = e^{\int q(y) dy}.$$

If neither quotient has the required dependence, these two tests do not supply an integrating factor.

An integrating factor depending only on x

Example (Solve $2y \, dx + (x + 1/x) \, dy = 0$, $x > 0$)

Here $M = 2y$, $N = x + 1/x$, and $M_y = 2 \neq 1 - 1/x^2 = N_x$: not exact. But

$$\frac{M_y - N_x}{N} = \frac{1 + 1/x^2}{x + 1/x} = \frac{1}{x}.$$

Thus $\mu(x) = e^{\int (1/x) \, dx} = x$ for $x > 0$. Multiplying by x gives

$$2xy \, dx + (x^2 + 1) \, dy = 0.$$

Integrating $\Phi_y = x^2 + 1$ with respect to y gives

$$\Phi(x, y) = (x^2 + 1)y + h(x).$$

Requiring $\Phi_x = 2xy$ gives $h'(x) = 0$. Thus

$$(x^2 + 1)y = C, \quad x > 0.$$

If $y(1) = 2$, then $C = 4$ and $y = 4/(x^2 + 1)$.

An integrating factor depending on y

Example (Solve $(2x + y^2) dx + (x^2 + xy^2 + 2xy) dy = 0$, $x > 0$)

Here $M_y = 2y \neq 2x + y^2 + 2y = N_x$: not exact. For $x > 0$, $M = 2x + y^2 > 0$, and

$$\frac{N_x - M_y}{M} = \frac{2x + y^2}{2x + y^2} = 1,$$

Thus $\mu(y) = e^{\int 1 dy} = e^y$. Integrating $\Phi_x = e^y(2x + y^2)$ gives

$$\Phi(x, y) = e^y(x^2 + xy^2) + g(y).$$

Requiring $\Phi_y = e^y N$ gives $g'(y) = 0$. Hence

$$e^y(x^2 + xy^2) = C.$$

Note: The other methods we've discussed (separable, linear, Bernoulli, homogeneous nonlinear) do not apply directly to this example.

Practice problems: other first order methods

In [Trench, Chapter 2](#):

- §2.4, Bernoulli equations: Exercises 1–4, 7–11
- §2.4, homogeneous nonlinear equations: Exercises 15–18, 22–24
- §2.5, exact equations: Exercises 1–9, 18–20, 33–34
- §2.6, one-variable integrating factors: Exercises 3–7, 9–12

Test 1: action required

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